

Highlights

Label-free covariance alignment for longitudinal EEG decoding

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- A 24,181-parameter EEG decoder emitting class and calibrated confidence.
- Label-free in-network alignment reduces session drift by 52–84% in 15 of 15 subjects.
- Test-time adaptation can track the streaming class and erase the signal it protects.
- Two cohorts have opposite adaptation-rate optima, costing 0.06–0.09 to get wrong.
- The admissible rate band is computable from the recording protocol before fitting.

Label-free covariance alignment for longitudinal EEG decoding

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ABSTRACT

A decoder that drives a powered lower-limb exoskeleton is fitted once and must keep working for weeks, while electrode impedance, montage and skin contact change the second-order statistics that every spatial filter depends on. Re-estimating the input covariance at inference and whitening by it, without labels, corrects this shift inside the network and removes the need for a calibration session. Such a layer carries a failure mode its offline counterpart cannot have. A running estimate tracks whatever is currently streaming, and under a protocol that presents classes in long contiguous blocks it converges to the active class and whitens away the signal it was inserted to protect.

We present a 24,181-parameter decoder built around such a layer, and derive an admissible band for its adaptation rate from two timescales of the recording protocol. On a seven-participant exoskeleton cohort recorded across weeks it reaches 0.901 balanced accuracy with expected calibration error 0.039 (0.868 ± 0.026 over three seeds under the trace-normalised estimator), at 53 % of the parameters of the strongest baseline in the same pipeline. The layer reduces session-to-session covariance distance by 52.0 ± 4.5 % and 83.9 ± 3.9 % on two independent cohorts, in 15 of 15 participants ($p \leq 10^{-4}$), against a control with no systematic reduction. On the second cohort, whose class blocks last 309 windows, the decoder transfers at 0.792 without the layer, and enabling the layer costs 0.211. Slowing adaptation past the block length recovers 0.090 to 0.098 on every arm that uses alignment and exactly 0.000 on the arms that do not, while the same change costs 0.060 on the first cohort, so the two optima are opposite.

The band then predicted a third cohort in advance. With class blocks of five windows, both predictions committed before any model was trained held: enabling alignment changed accuracy by +0.002, and slowing it by -0.006, where the same slowing had recovered 0.090 on the second cohort. Retraining after ICA artefact removal lowers accuracy by 0.051 and leaves it far above chance, and the losses do not follow the amount of muscle activity removed. A trained selective head lowers accuracy in 14 of 18 arms and is reported as a negative result. Both bounds of the band are properties of the experimental design, so the condition can be checked before a model is trained.

1. Introduction

Spinal cord injury leaves roughly 250,000 to 500,000 people per year with impaired or absent lower-limb function, and powered exoskeletons driven by non-invasive brain signals are one route to restoring overground mobility [1–3]. Decoding the wearer's intent to walk or to stop from scalp EEG is the control problem at the centre of such systems [4, 5]. It is not well described by per-epoch classification accuracy alone, for three reasons that shape the design presented here.

Longitudinal drift The decoder is fitted once and must keep working weeks later. In the primary cohort used here this is literal: the model is fitted on sessions 1–3 and evaluated on sessions 4–9, recorded across weeks. Electrode montage, impedance and skin contact all change between sessions, and the resulting shift in second-order statistics is the dominant source of non-stationarity in EEG decoding [6–8]. Session drift degrades fixed spatial filters and is the reason most BCI systems require a per-session calibration block [9].

Asymmetric cost A false Walk command moves a person's legs. Balanced accuracy scores that identically to a missed Walk and cannot distinguish one sustained false activation from fifty scattered false windows. The quantity a wearer experiences is spurious activations per minute of standing, which we report alongside accuracy throughout.

Abstention A classifier trained only for accuracy must emit a decision for every window. A device that acts on a confidence value needs that value to be calibrated [10], and needs a principled way to do nothing when the evidence is weak [11, 12].

Existing responses to drift are mostly applied offline in preprocessing. Euclidean Alignment whitens each recording by the inverse square root of its mean covariance [13]; Riemannian Procrustes analysis and parallel transport generalise this to the manifold of symmetric positive-definite (SPD) matrices [14–16]. All of these estimate the alignment statistic over a complete recording, which means the target recording must exist before the decoder can run on it. Test-time adaptation removes that requirement in computer vision by updating normalisation statistics or parameters online [17–19], and recent work shows that such methods degrade when the test stream is temporally correlated instead of i.i.d. [20–22]. EEG data acquired under a blocked experimental protocol is temporally correlated by construction, which is the connection this paper makes precise.

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Our contributions are:

1. We introduce an adaptive alignment layer that estimates and applies a whitening transform without labels and continues to update at inference, and we verify that its output reaches the classifier instead of being undone by the following normalisation (Section 4.1).
2. We identify a failure mode specific to in-network alignment: the running estimate tracks the currently streaming class when class blocks exceed the estimator's memory, and the layer then removes the discriminative signal (Section 6.8).
3. We derive an admissible band for the adaptation rate, bounded below by the class-block timescale and above by the drift timescale, and show the condition is two-sided by testing it in the direction where we predicted no effect (Condition 1, Section 6.9).
4. We validate the mechanism separately from the task, reporting a $52.0 \pm 4.5\%$ and $83.9 \pm 3.9\%$ reduction in session-to-session covariance distance in 15 of 15 subjects against a no-op control (Section 6.1).
5. We report external validation on an independent cohort, where the decoder transfers and the alignment layer inverts the ordering of arms, and trace the inversion to the rate condition after controlling for sample size and implementation (Section 6.5).
6. We test the condition prospectively on a third cohort, with both predictions and their decision thresholds committed before any model was trained on it. Both predictions hold (Section 6.10).
7. We retrain the decoder after ICA artefact removal. Most of the accuracy survives, and the losses that remain do not follow the amount of muscle activity removed (Section 6.3).

We make no accuracy claim over published decoders. In the pipeline where both were evaluated, ATCNet [23] scores 0.913 against 0.901 for our configuration, and we report that as a loss in Section 6.4.

2. Related work

2.1. Compact convolutional decoders for EEG

ShallowFBCSPNet and Deep4Net established that a square $\rightarrow$ pool $\rightarrow$ log pipeline over learned spatial filters recovers the band-power features that filter-bank common spatial patterns computes explicitly [24]. EEGNet reduced the parameter count by an order of magnitude using depthwise-separable convolutions [25], and EEGNeX refined the same family [26]. These architectures work well within a session and are the standard baselines in the field [27, 28]. Their limitation for our setting is that the learned spatial filters are fixed at training time, so a montage or impedance change between sessions is not compensated at all. We respond by placing an adaptive whitening stage before the spatial filters, so that what the filters see is referred to a covariance estimated on the current recording (Section 4.1).

2.2. Attention-based decoders

ATCNet combines convolutional feature extraction, sliding-window multi-head attention and temporal convolution, and is the strongest baseline in our comparison [23]. EEG Conformer pairs a ShallowFBCSP-style stem with a transformer encoder [29], and EEG-TCNet applies temporal convolutions to the same problem [30]. The stem-plus-transformer combination is already published: an independent implementation of it in our pipeline reproduced EEG Conformer's score without improving on it (0.848 against 0.843), and we therefore claim no novelty for that pairing. What these models share with the convolutional family is a fixed front end, which is the property the present work targets.

2.3. Alignment and transfer in BCI

Riemannian methods treat the spatial covariance as a point on the SPD manifold and classify with geodesic distances [31–33]. For transfer, Euclidean Alignment whitens each subject or session by $\mathbf{M}^{-1/2}$ where $\mathbf{M}$ is its mean covariance [13]; Riemannian Procrustes analysis adds rotation and scaling [14], and parallel transport moves covariances between tangent spaces [15, 16]. These methods are effective and label-free, and reviews report them as the strongest general-purpose transfer tools for EEG [34, 35]. Their limitation is operational. The alignment statistic is computed over a whole recording, so the recording must be complete before decoding starts, and within-recording drift is not tracked. We respond by moving the estimator inside the network as a layer that updates online, and we then show what that move costs (Section 6.8).

2.4. Test-time adaptation

Updating normalisation statistics on the test distribution improves robustness to corruption in vision [18, 19], and Tent extends this to updating affine parameters by entropy minimisation [17]. Test-time training adds a self-supervised objective at inference [36]. A recurring finding in this literature is that these methods assume the test stream is i.i.d., and degrade when it is not: NOTE addresses temporally correlated streams by maintaining a class-balanced reservoir [20], SAR stabilises adaptation under distribution shift with small or imbalanced batches [21], and CoTTA and LAME address error accumulation in continual adaptation [22, 37]. Our contribution relative to this line is quantitative: we show that in a blocked EEG protocol the relevant comparison is between the estimator's effective memory and the duration of a single-class block, that this predicts the threshold location, and that the resulting constraint is two-sided because the same estimator is also what tracks the drift the layer exists to remove.

2.5. Calibration and selective prediction

Modern networks are overconfident, and expected calibration error [10, 38] degrades further under distribution shift [39], which is exactly the regime a longitudinally deployed decoder operates in. Selective classification allows a model to abstain [11, 12], and SelectiveNet trains the

classifier and the selection function jointly under a coverage constraint [40, 41]. We use SelectiveNet as published and claim no contribution to it; we include it because a decoder that must not move a wearer's legs on weak evidence needs the reject option, and because it is the only component in our ablation whose benefit transfers across both cohorts.

2.6. EEG correlates of gait and exoskeleton control

Walking modulates cortical activity in a gait-phase-locked manner [42, 43], and treadmill walking can be decoded from scalp EEG [5, 44]. Robot-assisted gait modulates midline sensorimotor rhythms [45]. The intent signal our decoder targets is event-related desynchronisation in the mu and beta bands, a power decrease over sensorimotor cortex preceding and accompanying movement [46, 47]. This motivates a representation in which log-power ratios are preserved (Section 4.2), because ERD is defined as a relative power change.

3. Problem formulation

Let $\mathbf{X} \in \mathbb{R}^{C \times T}$ be a window of C channels and T samples with label $y \in \{\text{Stop}, \text{Walk}\}$. A recording session s supplies windows drawn from $p_s(\mathbf{X}, y)$. We assume the conditional $p(y | \mathbf{X})$ is stable across sessions but the marginal $p_s(\mathbf{X})$ is not, which is the covariate-shift assumption standard in this literature [6, 48]. Concretely we take the shift to act primarily on the spatial covariance $\Sigma_s = \mathbb{E}_{p_s}[\mathbf{X}\mathbf{X}^\top]/T$.

The decoder is fitted on sessions S_{fit} and evaluated on disjoint later sessions S_{test} , with no labels available from S_{test} at any point. This is the constraint that rules out per-session recalibration and motivates a label-free correction.

We quantify session drift by the affine-invariant Riemannian distance between SPD matrices,

$$\delta(\mathbf{A}, \mathbf{B}) = \left\| \log(\mathbf{A}^{-1/2} \mathbf{B} \mathbf{A}^{-1/2}) \right\|_F, \quad (1)$$

in preference to the Frobenius distance, because covariances do not form a vector space and the Frobenius distance is dominated by overall scale, which would flatter any whitening operation [31, 33].

Two timescales govern the analysis in Section 6.8. Let τ_{blk} be the median duration, in windows, of a contiguous single-class block in the recording protocol, and let τ_{drift} be the timescale over which Σ_s changes appreciably. Both are properties of the experimental design and the recording hardware, independent of the model.

4. Method

The decoder maps a window to a class posterior and a selection score. Its four components are shown in Fig. 1 with exact parameter counts.

4.1. Adaptive alignment

The layer maintains a running mean spatial covariance $\mathbf{M}$, applies $\mathbf{M}^{-1/2}$, and interpolates between the aligned and

the raw signal through a learned scalar gate $g \in [0, 1]$:

$$\mathbf{X}' = g \mathbf{M}^{-1/2} \mathbf{X} + (1 - g) \mathbf{X}. \quad (2)$$

Each window's covariance is normalised by its own trace before being accumulated, so that the estimate is an average of shapes instead of an average weighted by power:

$$\mathbf{M} \leftarrow (1 - m) \mathbf{M} + \frac{m}{N} \sum_{n=1}^N \frac{\mathbf{C}_n}{\text{tr}(\mathbf{C}_n)/C}, \quad \mathbf{C}_n = \frac{\tilde{\mathbf{X}}_n \tilde{\mathbf{X}}_n^\top}{T}, \quad (3)$$

where $\tilde{\mathbf{X}}_n$ is the mean-centred window, N the batch size and m the momentum. Without the trace normalisation the estimate is power-weighted. On a cohort whose majority class carries 1.52× the power of the minority class, the unnormalised estimate sits at Riemannian distance 0.306 from the majority-class covariance, that is, it is the majority class; the normalised estimate sits at 3.899 and 3.194 from the two class covariances, between them. Section 6.7 reports what correcting this did and did not achieve.

Two properties separate this layer from batch normalisation [49]. The update in (3) uses only the covariance of incoming windows and therefore runs on unlabelled test data. And the estimate continues to update in evaluation mode, whereas BatchNorm freezes its running statistics at test time. Freezing is correct when train and test share a distribution; here they are weeks apart, and freezing discards the adaptation the layer exists to provide.

Propagation of the aligned signal A whitening stage placed before a normalisation layer invites the objection that the normalisation simply undoes it. With identical initialisation on held-out windows, stem outputs computed with and without alignment differ by 36 % in relative magnitude and their correlation structure differs by 0.108, so the transform is not absorbed downstream.

4.2. Multi-scale power stem

Three parallel temporal resolutions (64, 128 and 256 ms), each with its own depthwise spatial filters, are followed by square $\rightarrow$ pool $\rightarrow$ log, after Schirrmeister et al. [24]. Amplitude normalisation precedes the squaring so that log-power ratios survive: scaling a window by s shifts its log-power by $2 \log s$, a constant that a later affine layer absorbs, whereas normalising after the square destroys the ratio in which ERD is defined [46].

This ordering is load-bearing, and the evidence is a dose-response across representations that differ in how much marginal power they carry (Table 5). Removing per-window amplitude normalisation is worth +0.134 and +0.174 on band-power features, which are pure marginal power; −0.016 and +0.029 on tangent-space features, which carry power only through the covariance diagonal; and −0.004 and +0.000 on a correlation-only representation, which is scale-free by construction and therefore cannot respond. A representation that carries no marginal power shows no effect, which is what identifies the mechanism instead of merely demonstrating it.

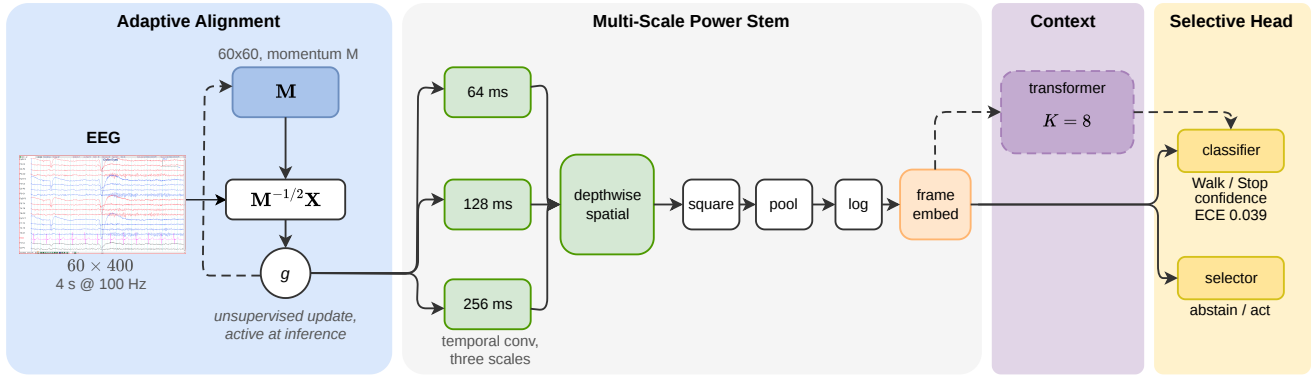


Figure 1: The decoder. The adaptive alignment layer maintains a running spatial covariance and whitens by it; its update path (dashed, left) uses no labels and remains active at inference, which is what separates it from a normalisation layer. The cross-epoch context pathway is optional and accounts for 594,816 of the 618,997 parameters of the full configuration, so the default configuration omits it and uses 24,181. Parameter counts are exact and sum to the total.

These are participant-paired comparisons, so they can be tested as such. Wilcoxon signed-rank tests over participants, Holm-corrected across the three representations within each cohort, give band power +0.174 (better in 7 of 8, $p_{\text{Holm}} = 0.047$) on cohort B and +0.134 (6 of 7, $p_{\text{Holm}} = 0.094$) on cohort A; tangent space +0.029 (5 of 8, $p_{\text{Holm}} = 0.500$) and -0.016 (2 of 7, $p_{\text{Holm}} = 0.750$); and the correlation-only control +0.000 (3 of 8, $p_{\text{Holm}} = 1.000$) and -0.004 (3 of 7, $p_{\text{Holm}} = 0.750$). **The effect reaches significance on the larger cohort and is null in the scale-free control, as the mechanism predicts;** cohort A is consistent in sign but underpowered at $n = 7$, where the smallest attainable Holm-corrected p is 0.047.

A separate sweep over where normalisation is applied in the network confirms the ordering at the architecture level: on cohort A with 4 s windows, per-window normalisation scores 0.849, global per-recording normalisation 0.878, and no normalisation 0.864, so the global setting used here is the best of the three and per-window normalisation is the worst.

4.3. Causal cross-epoch context

A causal transformer encoder [50] over a sequence of $K = 8$ epochs at stride 3 (a 14.5 s span), following the epoch-sequence formulation standard in sleep staging [51–53]. All masks are causal, so only present and past information is used, which is the only variant a worn device can run. Measured state persistence in cohort A supports a sequence model: the self-transition probability is 0.984 for Stop and 0.955 for Walk at a 0.5 s step, corresponding to dwell times of 12 to 32 s.

4.4. Selective head

A classifier paired with a selection function trained under a coverage constraint [40], with a full-coverage auxiliary term so that the backbone continues to learn from every sample, including the declined region. We use the formulation as published.

5. Experimental setup

5.1. Data

Cohort A (exoskeleton, OpenNeuro ds007788). Seven participants, nine sessions each, recorded across weeks while walking in a powered lower-limb exoskeleton [4]. Sixty-channel active EEG at 100 Hz; walk/stop labels are taken from the exoskeleton controller state. Windows of 4 s at 0.5 s stride give approximately 5031 fitting windows per participant. The model is fitted on sessions 1–3 and evaluated on sessions 4–9.

Cohort B (treadmill MoBI). Eight participants, three trials each, 64-channel EEG at 100 Hz, walk/stand from treadmill phase [44]. Fitted on trial 1 and evaluated on trials 2–3, with 2 s windows. The cohort is from an independent laboratory with different sensors, a different task framing, and reversed class balance: Stop is the minority class at approximately 12 % here and the majority class in cohort A.

Cohort C (motor execution, PhysioNet EEGMIDB). Twenty participants, S001–S020, from the PhysioNet motor movement and imagery database [54, 55], recorded with 64-channel EEG at 160 Hz and resampled to 100 Hz. Only the motor-execution runs are used, and the task is rest against movement of the hands or feet. The model is fitted on runs 3, 5, 7 and 9 and evaluated on runs 11 and 13, with 2 s windows at 0.5 s stride, each lying wholly inside one annotated period. The protocol alternates the two classes about every 4 s, a median of 5 windows per class block. Its analysis plan was committed to version control before any model was trained on it (Section 6.10).

The protocols differ sharply in temporal structure, which Section 6.8 shows to be decisive (Table 1).

5.2. Preprocessing and training

Signals are band-pass filtered to 8–30 Hz and resampled to 100 Hz. Cohort A uses a zero-phase FIR filter in MNE-Python [56], after which the EOG channels, filtered to the same band, are regressed linearly out of the EEG. Cohort B uses a zero-phase fourth-order Butterworth filter and has no EOG step. No other artefact rejection, ICA decomposition

Table 1

Contiguous single-class block structure. τ_{blk} is the median block duration in windows.

Cohort	Blocks	τ_{blk}	Longest
A (exoskeleton)	109	18	126
B (treadmill)	7	309	2211

Table 2

Training configuration, identical across all arms and both cohorts except where the cohort definition differs.

Setting	Value
Band-pass / resample	8–30 Hz, 100 Hz (zero-phase)
Artefact correction	EOG regression (A); none (B)
Amplitude normalisation	global, per recording
Window / stride	4 s / 0.5 s (A), 2 s (B)
Context length K , stride	8 epochs, 3 (14.5 s span)
Optimiser	AdamW [57], lr 3×10^{-4} , weight decay 10^{-2} , one-cycle schedule
Batch size N	32
Epochs	at most 30; early stopping after 12 without improvement
Gradient clipping	norm 1.0
Loss	cross-entropy, inverse-frequency weighted
Selective coverage target	0.90
Adaptation momentum m	0.2 (swept in §6.8)
Model selection	best held-out balanced accuracy

or channel interpolation is applied, so that the alignment layer is evaluated on the shift it is meant to correct. The band also excludes the gait cycle, near 1–2 Hz, and most electromyographic power above 30 Hz. Amplitude normalisation is applied globally per recording, before the squaring stage, for the reason given in Section 4.2.

Table 2 lists the training configuration. The adaptation momentum is $m = 0.2$ unless stated otherwise; Section 6.8 treats m as the object of study. Model selection uses balanced accuracy on a held-out split of the fitting sessions, evaluated at the best epoch by that criterion, which is conventional but upward-biased on small folds (Section 8).

Backbone screening Before settling on the stem of Section 4.2 we screened 19 published decoders from the *braindecode* library [58] under identical preprocessing (the *Screen* column of Table 3). Sixteen converged, spanning 0.500 to 0.766; four did not.

The screen was first run on three participants, and repeating it on all seven changed the answer. Across the fifteen models present in both runs the mean absolute change is 0.023 and the largest is 0.061, and **eleven of fifteen models change rank**. The first-ranked model changes from TSception (0.788 on three participants) to FBLightConvNet (0.766 on seven). The sharpest case is EEG Conformer [29], which ranked first on three participants at 0.827 and fails to converge on seven. Selecting a backbone from the three-participant screen would therefore have selected a model that

does not run on the full cohort. A cheap measurement that ranks models is not a cheap version of an expensive one (Section 5.4).

On the four failures. ATCNet cannot be run in this harness at all: its final layer is a `ModuleList` that the backbone adapter cannot consume, which is why the original sweep excluded it. The other three (EEG Conformer, EEG-ITNet, FBMSNet) raise a not-a-number error during training. We verified that this is not a data fault: none of the seven participants' epoch tensors contains a not-a-number or infinite value. These are training divergences under this configuration and are recorded as such, providing no evidence against those models.

5.3. Outcome measures

Balanced accuracy for comparability with the literature; expected calibration error (ECE) with 10 equal-width bins [38], because a device acting on a confidence value needs that value to mean something; accuracy at 90 % coverage, the trained selective operating point; spurious activations per minute of standing; and detection latency, reported alongside the benefit it buys.

Leakage guard Unconditioned nuisance-decodability is confounded with accuracy. In our pipeline a decoder that uses no movement information whatever measures $R^2 = +0.863$ against the movement signal, because both the decoder output and the nuisance are driven by the class label. Movement leakage is therefore measured conditional on the class label throughout, by centring the nuisance features within class before regression.

We calibrate the probe by injecting the recorded movement signal into the EEG, spatially mixed with a fixed per-participant gain matrix, at a known fraction f of signal power. The calibration uses the tangent-space representation. Accuracy rises monotonically with f : 0.763 at $f = 0$, then 0.819, 0.873, 0.892 and 0.913 at $f = 0.001, 0.005, 0.01$ and 0.05 . The conditional probe rises with it, from -0.112 at $f = 0$ through $+0.015$ at $f = 0.005$ to $+0.067$ at $f = 0.02$, where accuracy is 0.903. Reaching an accuracy near 0.90 through injected movement therefore moves the probe to about $+0.07$. The proposed *align + gate* decoder reaches 0.901 with a conditional probe of -0.068 , and all eight cohort-A arms lie between -0.066 and -0.082 , on the uncontaminated side of the calibration curve. Two limits apply. The calibration was performed on a different representation from the decoder it is compared with, and the conditional probe cannot register artefact information that is constant within each class. The comparison is therefore evidence against movement-driven accuracy. Section 6.3 adds a second, independent test by removing artefact components and retraining.

5.4. Statistical treatment and measurement noise

Drift reductions are tested with a two-sided paired t -test over participants, pairing each participant's aligned distance against its raw distance. Accuracy comparisons are reported

as the mean over participants with the between-subject standard deviation; we do not perform paired subject-level significance tests on the accuracy comparisons, and Section 8 states why this matters.

Two ablation arms disable alignment entirely and therefore never invoke the running estimator. For them the momentum and the estimator version are inert, so every run is a true repeat and they measure the noise floor directly (Table 9).

The stem-only arm reproduces 0.7375 bit-for-bit across three runs; the arms containing the cross-epoch transformer do not, and the attention path is the only structural difference between them. We attribute the variation to non-deterministic reductions in scaled dot-product attention on GPU.

Run-to-run variation at a fixed split seed is a lower bound on the relevant noise. Repeating the two strongest arms at a second split seed (Table 9) gives a between-seed spread of 0.012 to 0.020, about twice the fixed-seed figure. The published baselines are worse: over three seeds in the same pipeline, ShallowFBCSPNet spans 0.053 and ATCNet 0.041 (Table 7). This is the quantity that governs whether two models can be ranked, and it is why no single-seed ranking is reported in this paper.

Three consequences follow. Any arm containing the transformer carries approximately 0.010 run-to-run noise at fixed seed and up to 0.020 across seeds, so differences below 0.02 are not interpreted anywhere in this paper. The effects we do interpret (+0.090 to +0.098 in Table 4, -0.060 in Table 4) are about 2.3 to 3.8 times the standard deviation across three data-split seeds. And an exact null in the deterministic arm is stronger evidence than one merely within noise, because that arm has no noise to hide in.

Pipeline separation Our architecture and the ATCNet variants were evaluated in one training pipeline; the remaining published baselines were evaluated in an earlier one. ATCNet was run in both and scores 0.913 and 0.881 respectively, so the two pipelines differ by 0.032 for an identical model, which is larger than most differences of interest in this paper. We therefore report the two groups separately and never compare across them.

6. Results

6.1. The drift mechanism

We measure the Riemannian distance (1) between the fitting-session mean covariance and each held-out session's, before and after the layer (Fig. 2, Table 8).

Drift is reduced in 15 of 15 participants ($t = -9.14$, $p = 1 \times 10^{-4}$ for cohort A; $t = -20.59$, $p < 1 \times 10^{-5}$ for cohort B), with per-participant reductions spanning 45.2 to 56.1 % on cohort A and 76.3 to 88.6 % on cohort B (Table 8). No participant regresses, and the reduction is not carried by one or two outliers. The no-op control shows no systematic reduction in either cohort, which establishes that the effect comes from adaptation. The control excludes the whitening operation and the measurement itself.

The control is not inert. Per participant it moves the distance by -28.8% to $+45.0\%$, scattering in both directions about a mean of roughly zero. Two implementation facts explain this. First, the running estimate is primed by copying the first batch's covariance regardless of momentum, so at momentum 0 the layer does not whiten by the identity. It whitens by the fit-session covariance and then never updates. Second, the layer returns a gated blend $gWx + (1 - g)x$ with $g = \sigma(2.0) \approx 0.88$, so the effective operation is the fixed linear map $M = gW + (1 - g)I$. A fixed map applied to both covariances is a congruence, under which the affine-invariant distance is invariant; but the measurement normalises each covariance by its own trace *after* the map, and M scales the fit and held-out traces by different factors. The normalised pair is therefore no longer related by a congruence, and the residual scatter is that trace renormalisation. The eigenvalue clamp at 10^{-6} contributes on near-singular montages. The control is still the right one, since it isolates adaptation from whitening, but its scatter measures the noise floor of the measurement itself.

The mechanism works as designed, and Section 6.5 shows that this does not settle whether it helps the task.

6.2. Component ablation, cohort A

The acc@90 value for the *align + ctx* arm is withheld because with the selection head disabled the model emits near-constant confidence, so ranking by it selects the first 90 % of the test set in recording order. That number would be an ordering artefact.

Selective prediction Abstention is supposed to raise accuracy on the windows that are kept. It does not. Across the 18 arms in this work for which both quantities were computed on the same windows with the same metric, declining the least-confident 10 % **lowers** balanced accuracy in 14, by 0.045 on average; the headline arm loses 0.026 (0.901 $\rightarrow$ 0.876) and the worst case loses 0.122. The four arms where abstention helps are exactly those with no trained gate (*align + ctx, stem only*), where the ranking is arbitrary. A gate that is beaten by its own absence is not ranking windows by correctness, and the selective head is therefore reported as a component for which no benefit was demonstrated. Fixing it, for instance by thresholding a temperature-scaled softmax confidence instead of the learned gate, is left open; the calibration result (ECE = 0.039) is separate and stands.

The head contributes no accuracy either. Under the trace-normalised estimator, an *align*-only arm reaches 0.883 against 0.884 with the head, a difference of 0.002, and adding the head to *align + ctx* changes accuracy by 0.000 (0.862 in both arms). Calibration does not suffer either: without the head ECE is 0.038, against 0.042 with it, both under the same estimator. An earlier difference of +0.016 between the same two context arms under the original estimator is 1.6 $\times$ the 0.010 noise floor of arms containing the transformer (Table 9), and it does not reappear in the comparison made with one estimator throughout. On this single-seed evidence the selective head could be removed with no loss of accuracy or calibration. It is kept in the

Table 3

Every decoder evaluated in this work, one row per model. **Bold marks the best value in each column. The three accuracy columns are different experiments and must not be compared across:** they are three different training pipelines on the same 7 participants. ATCNet, the one model run in two of them, differs by 0.032 between Pipe. C and Pipe. F, which bounds how much of any cross-column gap is pipeline rather than model. — means the experiment was not run for that model; *n/a* means the model cannot produce that quantity, having no selective head. ATCNet leads on balanced accuracy by 0.012 over this architecture, which leads on parameter count, calibration and the abstention operating point. *Spread* is the range over seeds; at 0.053 it exceeds that 0.012 gap, so the gap is not resolved by the seeds we ran. Parameter counts are instantiated at this study's input shape, not quoted from the source papers. Screening implementations from braindecode [58]; "n/c" did not converge. ATCNet ablations are in Table 7, not here: this architecture is not an ATCNet derivative and shares no code with it.

Model	Params	Screen	Pipe. C	Pipe. F	ECE ↓	Acc@90	Spread
<i>This work</i>							
Ours, align + gate	24 181	—	0.901	—	0.039	0.876	0.019
Ours, align + ctx + gate	618 997	—	0.878	—	0.074	0.837	0.012
Ours, stem only	6 768	—	0.827	—	0.082	0.823	—
<i>Published decoders</i>							
ATCNet [23]	45 478	n/c	0.913	0.879	<i>n/a</i>	<i>n/a</i>	0.041
EEGNeX [26]	58 082	0.640	—	0.876	<i>n/a</i>	<i>n/a</i>	0.041
ShallowFBCSPNet [24]	98 802	0.680	—	0.867	<i>n/a</i>	<i>n/a</i>	0.053
CALMNet-bare	3 946	—	—	0.858	<i>n/a</i>	<i>n/a</i>	0.040
EEGConformer [29]	440 706	n/c	—	0.843	<i>n/a</i>	<i>n/a</i>	0.018
FBLightConvNet	19 000	0.766	—	—	<i>n/a</i>	<i>n/a</i>	—
EEGInceptionMI	3 708 002	0.764	—	—	<i>n/a</i>	<i>n/a</i>	—
TSception [59]	165 074	0.760	—	—	<i>n/a</i>	<i>n/a</i>	—
MSVTNet	73 468	0.751	—	—	<i>n/a</i>	<i>n/a</i>	—
EEGSimpleConv	768 954	0.702	—	—	<i>n/a</i>	<i>n/a</i>	—
FBCNet [60]	22 000	0.691	—	—	<i>n/a</i>	<i>n/a</i>	—
SincShallowNet [61]	7 810	0.678	—	—	<i>n/a</i>	<i>n/a</i>	—
SCCNet [62]	7 228	0.646	—	—	<i>n/a</i>	<i>n/a</i>	—
IFNet [63]	15 730	0.615	—	—	<i>n/a</i>	<i>n/a</i>	—
Deep4Net [24]	281 577	0.562	—	—	<i>n/a</i>	<i>n/a</i>	—
EEGNet [25]	2 450	0.560	—	—	<i>n/a</i>	<i>n/a</i>	—
CTNet [64]	152 282	0.542	—	—	<i>n/a</i>	<i>n/a</i>	—
BDTCN	33 752	0.510	—	—	<i>n/a</i>	<i>n/a</i>	—
EEGTCNet [30]	4 886	0.500	—	—	<i>n/a</i>	<i>n/a</i>	—
FBMSNet [65]	26 419	n/c	—	—	<i>n/a</i>	<i>n/a</i>	—
EEGITNet [66]	3 180	n/c	—	—	<i>n/a</i>	<i>n/a</i>	—

arms reported here so that they remain comparable with the ablation.

Dependence on the alignment stage With alignment removed, the context and selection modules together contribute between 0.000 and 0.007: the no-align arm spans 0.827–0.834 across repeats (Table 9) while the stem-only arm is deterministic at 0.827. Against the 0.010 noise floor that is indistinguishable from nothing, and well below alignment's own contribution: +0.051 under the original estimator (0.878 against 0.827, Table 10) and +0.028 under the trace-normalised one (0.862 against 0.834) from the estimator-fix run of Section 6.7. The lower figure should be

read with care: the no-align arm is inert to the estimator, so its 0.834 differs from 0.827 only by run-to-run variation, and part of the gap between the two figures is that noise. The comparison that isolates alignment most directly, *align + gate* against *gate* with nothing else changed, was repeated over three data-split seeds under the trace-normalised estimator. With each participant's accuracy averaged over seeds, alignment raises accuracy by +0.026 (0.868 against 0.842), higher in 6 of 7 participants (Wilcoxon $p = 0.22$). The effect is +0.067, +0.029 and −0.019 on the three seeds, and it is uneven across participants: sub-04 gains 0.179 and sub-05 loses 0.127, each with the same sign on every seed. The gain

Table 4

The adaptation-rate condition, single seed. Momentum m sets the estimator's memory to roughly $32/m$ windows: 160, 640 and 3200. **Every arm that uses alignment gains as the memory crosses the class-block length, and neither arm that omits it responds** — the stem-only arm reproduces 0.737 exactly at all three settings despite a 160-fold change. The lower block applies the identical change to the cohort whose blocks are short, where it costs 0.060: the optima are opposite and no single rate serves both protocols.

Arm	$m=0.20$	$m=0.05$	$m=0.01$	Δ at crossing	Aligns	Δ cohort A
<i>Cohort B ($\tau_{\text{blk}} = 309$): slowing adaptation past the class block</i>						
align + ctx + gate	0.631	0.724	0.723	+0.093	yes	—
align + gate	0.581	0.671	0.696	+0.090	yes	—
align + ctx	0.606	0.700	0.726	+0.094	yes	—
ctx + gate	0.788	0.782	0.783	-0.007	no	—
stem only	0.737	0.737	0.737	+0.000	no	—
<i>Cohort A ($\tau_{\text{blk}} = 18$): the same change, matched estimator</i>						
align + ctx + gate	0.862	—	0.801	-0.060	yes	-0.060
ctx + gate	0.834	—	0.827	-0.007	no	-0.007

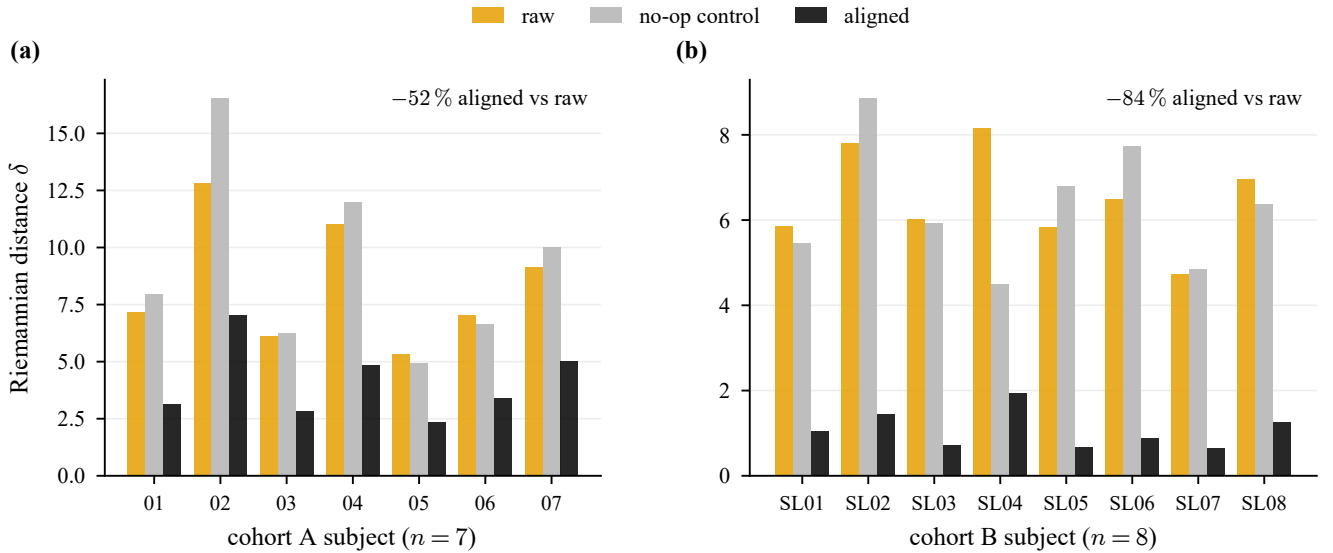


Figure 2: Session-drift reduction per participant. Bars are the Riemannian distance δ between the fitting-session mean covariance and the held-out session mean, for the raw signal, the momentum-0 no-op control, and the aligned signal. (a) cohort A, $n = 7$; (b) cohort B, $n = 8$. Distance falls in every participant of both cohorts; the no-op control, which shares every code path but never updates its estimate, shows no reduction.

from alignment on cohort A is therefore close to the trace-normalised single-seed figure of +0.028, and with seven participants it is not significant. The modules do not stack; they depend on the alignment stage to be useful at all.

Cross-epoch context Adding it costs 0.023 accuracy and raises ECE from 0.039 to 0.074, while reducing spurious activations from 1.64 to 1.26 per minute, a 23 % reduction. It is therefore reported as a selectable safety mode, outside the default configuration. Removing it also removes 594,816 of the model's 618,997 parameters, which is what makes the default configuration compact.

6.3. Artefact control

Walk and stop differ in movement, so accuracy on this task could arise from muscle or motion artefact. The proposed configuration was therefore retrained after independent component analysis, with both arms restricted to the training task so that they differ only in the cleaning step. Each recording was re-referenced to the average, a copy band-passed to 1–45 Hz was decomposed into 30 components with the extended Picard algorithm [67], and the components that ICLabel [68] classified as eye, muscle, heart, line noise or channel noise were removed before band-passing to the decoding band. Across the 63 recordings, a mean of 6.8 of 30 components was removed, most often eye (42 % of removals) and muscle (38 %). Gait-related artefact is difficult to separate from cortical activity during walking [69, 70],

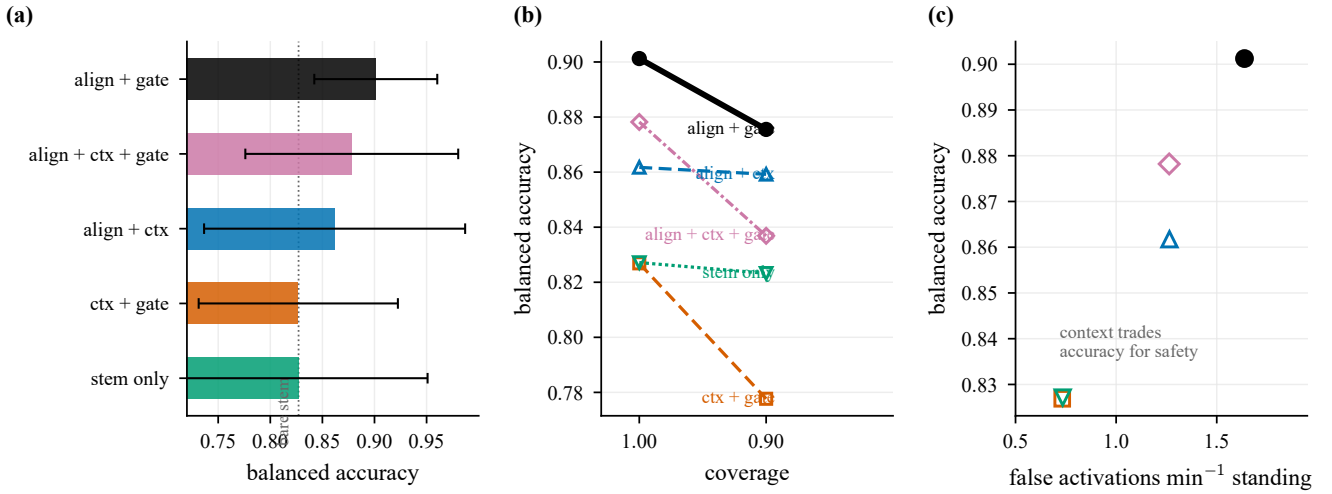


Figure 3: Component ablation on cohort A, seed 0. (a) Balanced accuracy by arm; error bars are ± 1 SD across the 7 participants; seed variation is reported separately in Table 9. (b) Balanced accuracy at 90% coverage, obtained by declining the least-confident 10% of windows. It sits *below* full-coverage accuracy for every arm with a trained gate. (c) Balanced accuracy against spurious activations per minute of standing, showing the cross-epoch context as a selectable operating point.

Table 5

Two-condition comparisons. **The first block is the identifying evidence for the stem design:** removing per-window normalisation is worth +0.134 and +0.174 where the representation is pure marginal power, and **exactly +0.000 where it is scale-free by construction** and therefore cannot respond. An effect that scales with power content and vanishes without it identifies the mechanism rather than merely demonstrating it. The second block retired an amplitude side-channel: supplying a per-window amplitude feature and supplying the same feature with its window assignment shuffled differ by under 0.02 for four of five models.

Comparison	Power content	Condition	Control	Δ
<i>Per-window amplitude normalisation, by how much marginal power the representation carries</i>				
band power, cohort A	full	0.625	0.760	+0.134
band power, cohort B	full	0.574	0.748	+0.174
tangent space, cohort A	partial	0.762	0.746	-0.016
tangent space, cohort B	partial	0.675	0.704	+0.029
correlation, cohort A	none	0.738	0.734	-0.004
correlation, cohort B	none	0.701	0.702	+0.000
<i>Amplitude side-channel against its shuffled control</i>				
ShallowFBCSPNet	—	0.832	0.816	-0.017
Deep4Net	—	0.798	0.809	+0.011
EEGNet	—	0.841	0.841	-0.000
EEGITNet	—	0.828	0.837	+0.008
BDTCN	—	0.748	0.652	-0.095

and ICLabel is not trained on it specifically, so this control narrows the question and leaves part of it open.

Cleaning lowers balanced accuracy from 0.835 to 0.785, a paired difference of -0.051 (higher in 2 participants, lower in 5; Wilcoxon $p = 0.22$; Table 11). Most of the accuracy therefore survives, well above the 0.500 chance level, and four participants are essentially unchanged or improve. The loss is concentrated in three participants, of 0.11 to 0.16. It does not follow the amount of muscle activity removed. The participant with the most muscle components removed, 7.1 per recording, improves by 0.016, and one with 0.3 removed loses 0.124 (Spearman $\rho = +0.22$ between muscle components removed and accuracy change). Dependence on

Table 6

Where amplitude normalisation is applied, across seven published decoders on cohort A, single seed. All three columns are balanced accuracies, not changes; best per row in bold. **Per-window normalisation is best for none of the seven and worst for four**, which is the architecture-level counterpart of the representation-level result in Table 5 and the reason the stem normalises globally before squaring.

Model	Per-window	Global	None
ShallowFBCSPNet [24]	0.824	0.821	0.818
Deep4Net [24]	0.760	0.821	0.821
EEGNet [26]	0.834	0.832	0.835
EEGITNet [66]	0.833	0.791	0.795
BDTCN	0.820	0.822	0.820
EEGConformer [29]	0.828	0.829	0.805
ATCNet [23]	0.821	0.828	0.843

muscle artefact would produce the opposite pattern. Two of the three largest losses fall on the participants with the lowest baseline accuracy, 0.673 and 0.680, where estimates are noisiest. The conditional leakage probe stays negative in both arms (-0.088 and -0.077). With seven participants, and with re-referencing also part of the cleaning, the control falls short of establishing that the accuracy is neural, and it gives no support to the view that the accuracy rests on muscle artefact.

6.4. Comparison with published decoders

In the pipeline where both were measured, ATCNet scores 0.913 against 0.901 at seed 0 and 0.892 for our two-seed mean (Table 9). The gap of 0.012 to 0.021 sits inside the 0.020 between-seed spread of Section 5.4, so two seeds cannot separate the two models. It is reported as a loss, since the point estimate favours ATCNet in both seeds we ran.

Table 7

Every architectural intervention tested, with the control that decided it. The motion-canceller row is the clearest rejection: the module scored 0.905 with the inertial channel present and chance with it zeroed, so it was an inertial classifier rather than an EEG one. The temporal-smoothing rows are the clearest retention: shuffling window order does not merely remove the gain, it drops **0.113 below the unsmoothed baseline**, which no operation helping for an unrelated reason would do. n is seeds.

Intervention	n	With	Control	Δ	Verdict
<i>Rejected: the control removed the effect</i>					
Motion cancellation	1	0.905	0.501 (zeroed)	-0.404	inertial classifier
Motion cancellation	1	0.905	0.529 (shuffled)	-0.376	same
Within-epoch attention	3	0.840	0.863 (removed)	+0.024	block not earning its place
MRCP dual-band pathway	1	0.742 (both)	0.775 (ERD only)	+0.034	low band adds noise
Capacity scaling	1	0.768	0.608	+0.159	24 configs, no trend
Alignment (SPD backbone)	3	0.823 (EA)	0.821 (none)	-0.002	inside seed spread
<i>Kept: the control preserved the effect</i>					
HMM forward filter	4	0.802	0.778 (baseline)	+0.024	survives shuffle
Viterbi decoding	4	0.817	0.778 (baseline)	+0.038	survives shuffle
control: shuffled order	4	0.665	0.778 (baseline)	-0.113	falls below baseline

Table 8

Per-participant session drift. δ is the affine-invariant Riemannian distance between the fitting-session mean covariance and the held-out mean. The no-op control shares every code path but never updates its estimate. **Every participant improves:** 15 of 15, $t = -9.14$, $p = 10^{-4}$ (A) and $t = -20.59$, $p < 10^{-5}$ (B); the control shows no reduction in either cohort.

Participant	Raw δ	No-op	Aligned	Reduction
<i>Cohort A, 6 held-out sessions</i>				
A01	7.154	7.938	3.144	56.1 %
A02	12.832	16.522	7.013	45.3 %
A03	6.092	6.228	2.827	53.6 %
A04	11.035	11.981	4.840	56.1 %
A05	5.336	4.906	2.346	56.0 %
A06	7.016	6.644	3.399	51.5 %
A07	9.139	9.990	5.007	45.2 %
<i>Cohort B, 2 held-out trials</i>				
SL01	5.845	5.450	1.039	82.2 %
SL02	7.795	8.850	1.451	81.4 %
SL03	6.030	5.934	0.713	88.2 %
SL04	8.158	4.486	1.933	76.3 %
SL05	5.823	6.799	0.666	88.6 %
SL06	6.482	7.722	0.888	86.3 %
SL07	4.738	4.849	0.653	86.2 %
SL08	6.966	6.378	1.250	82.1 %

Transfer to an alternative backbone If the selection and context modules were generally useful, attaching them to a stronger backbone should help. They do not: stock ATCNet with our cross-epoch context and selective head scores 0.894 against 0.913 for stock ATCNet alone, a loss of 0.018 (Table 3). The modules improve calibration (ECE 0.058, and an abstention point at 0.849) and cost accuracy, which is the same trade seen within our own architecture in Table 10: accuracy is exchanged for calibration.

What the configuration offers instead is parity with the strongest baseline at 53 % of its parameters and 5 % of EEG

Table 9

Measurement noise, and why no single-seed ranking appears in this paper. The first block repeats arms for which momentum and estimator version are inert, so all variation is run-to-run: the arm without the cross-epoch transformer is bit-identical across three runs, and the arms with it are not. Across split seeds the spread doubles, and for the published baselines it reaches 0.053. **Differences below 0.02 are not interpreted anywhere in this work.**

Arm	Transformer	n	Spread
<i>Fixed seed, repeated runs of arms where the manipulated variable is inert</i>			
stem only, cohort B	no	3	0.0000
no-align, cohort A	yes	3	0.0069
no-align, cohort B	yes	4	0.0107
<i>Across split seeds, our arms</i>			
align + gate	yes	2	0.0195
align + ctx + gate	yes	2	0.0119
<i>Across split seeds, published baselines</i>			
ShallowFBCSPNet	—	3	0.0534
ATCNet	—	3	0.0413

Table 10

Component ablation, cohort A, single seed. Accuracy is the mean over 7 participants with the between-subject SD. Acc@90 is accuracy at 90 % coverage; FA/min is spurious activations per minute of standing. All cells use the original estimator so that arms are mutually comparable.

Components	Acc $\pm$ SD	ECE	Acc@90	FA/min
align + gate	0.901 $\pm$ 0.059	0.039	0.876	1.64
align + ctx + gate	0.878 $\pm$ 0.102	0.074	0.837	1.26
align + ctx	0.862 $\pm$ 0.125	0.072	—	1.26
ctx + gate	0.827 $\pm$ 0.096	0.091	0.778	0.73
stem only	0.827 $\pm$ 0.124	0.082	0.823	0.73

Conformer's (Fig. 4), together with calibrated confidence (ECE = 0.039) and a trained abstention operating point (0.876 at 90 % coverage), neither of which the baselines provide. The baselines were not compressed for this comparison, so the parameter axis reports each architecture at its

Table 11

Artefact control on cohort A. Balanced accuracy of the proposed configuration on the training task, before and after ICA cleaning, with the number of muscle components removed per recording.

Participant	Uncleaned	Cleaned	Change	Muscle removed
sub-01	0.969	0.955	-0.014	2.9
sub-02	0.898	0.914	+0.016	7.1
sub-03	0.955	0.841	-0.114	2.3
sub-04	0.835	0.817	-0.018	1.9
sub-05	0.839	0.893	+0.053	1.3
sub-06	0.673	0.548	-0.124	0.3
sub-07	0.680	0.525	-0.155	2.3
Mean	0.835	0.785	-0.051	

Table 12

Both cohorts, single seed, original estimator.

Components	Cohort A	Cohort B
align + gate	0.901	0.581
align + ctx + gate	0.878	0.626
align + ctx	0.862	0.606
ctx + gate	0.827	0.792
stem only	0.827	0.737

published size, above the minimum it could be compressed to.

6.5. External validation and the inverted ordering

The decoder transfers: without the alignment layer it reaches 0.792 on cohort B, against a majority-class floor of 0.500 balanced accuracy on a cohort whose classes split 87.7/12.3. What does not transfer is the alignment layer. The best cohort-A configuration is the worst cohort-B one, and the ordering inverts, with all three arms that use alignment falling below both arms that do not. The layer is worth +0.074 on cohort A (0.901 against 0.827) and -0.211 on cohort B (0.581 against 0.792). The cohort A figure comes from one seed and also removes the context module. Against the gate-only arm over three seeds, alignment is worth +0.026 there (Section 6.2), so the size of the inversion comes mostly from the loss on cohort B. The remainder of the results characterises this inversion, which proves more transferable than the architecture itself.

A protocol-level switch The two cohorts do not require two architectures. The admissible band of Condition 1 is a switch whose position is fixed by the recording protocol: with the default estimator memory $\mu = 32/m = 160$ windows, cohort A's $\tau_{\text{blk}} = 18$ satisfies $\tau_{\text{blk}} < \mu$ and cohort B's $\tau_{\text{blk}} = 309$ does not. Enabling alignment exactly when the band is satisfied selects 0.901 on cohort A and 0.792 on cohort B: **the best available configuration on each**, from one model and one rule evaluated before any training.

Learnability of the switch The architecture already contains a learnable version of it: the blend weight $g = \sigma(\text{gate})$

in (3) is a trained parameter, so the model is free to close the gate and bypass alignment. On cohort B it does not, and scores 0.626. The reason is structural and does not lie in optimisation. The gate is trained on the fitting sessions, where the estimator sees interleaved windows and alignment is harmless; the class-tracking failure appears only at test time, when the estimator meets contiguous blocks longer than its memory. **No training signal contains the pathology, so no gradient can close the gate against it.** This is the argument for a protocol-level check instead of an end-to-end one: the quantity that decides whether the layer helps is a property of the evaluation stream, which the fitting distribution does not expose.

6.6. Interventions tested and rejected

Six architectural interventions were tested before the configuration reported here and rejected by their own controls (Table 7). We report them because each control is what distinguishes a mechanism from a correlation, and because two of them looked convincing until the control was run.

Three are multi-seed and test a component the architecture could plausibly have kept.

Temporal smoothing and the attention block Applying an HMM forward filter over the decoder's output improves cohort A by +0.024 over four seeds, and Viterbi decoding by +0.038 (Table 7). Viterbi is reported for reference only: it needs the complete sequence and so cannot run in the causal setting this decoder targets. The control applies the same filter after shuffling window order, which preserves the operation and destroys the temporal structure: it falls to 0.665, **0.113 below** the unsmoothed baseline. A smoothing operation that helped for any reason other than genuine persistence would not behave that way. Participant-paired Wilcoxon signed-rank tests over the seven participants, with seeds averaged within participant and Holm correction across the three comparisons, place the forward filter at +0.024 (better in 6 of 7, $p_{\text{Holm}} = 0.063$), Viterbi at +0.038 (6 of 7, $p_{\text{Holm}} = 0.063$) and the shuffled control at -0.113 (worse in 7 of 7, $p_{\text{Holm}} = 0.047$). **Only the control clears 0.05:** with $n = 7$ the smallest attainable two-sided p is 0.016, so the smoothing gains are consistent in sign but underpowered, while the control's collapse is not. The cross-epoch pathway of Section 6.8 is therefore a real effect even though it costs accuracy.

By contrast, merging the power stem with a within-epoch attention block (Table 7) is improved by 0.024 when the attention is *removed*, across three seeds. The block was dropped.

Replication on an independent architecture We built a separate SPD-manifold decoder in which alignment can be fixed (Euclidean Alignment), learned, or omitted (Table 7). Over three seeds the three variants span 0.814 to 0.823, and omitting alignment entirely costs 0.002, far inside the ± 0.02 to ± 0.03 seed spread. A different backbone, a different alignment parameterisation, and the same conclusion: the alignment stage is not what carries the performance.

Normalisation across published architectures Table 6 repeats the normalisation comparison across seven published decoders. Per-window normalisation is the best choice for none of them and the worst for four, which is the architecture-level counterpart of the representation-level dose–response in Table 5. Table 5 reports the control that retired the amplitude side-channel: supplying a per-window amplitude feature and supplying the same feature with its window assignment shuffled differ by less than 0.01 for five of six models.

Tables 5 and 7 report the representation and architecture sweeps that preceded the design. The tangent-space representation is the strongest single feature family, which is what motivated a covariance-based alignment stage; and no structural change in the 24-configuration sweep reaches 0.06 over the baseline, with widening the network worth less than changing the readout.

The movement-related cortical potential (MRCP) pathway deserves a sentence because it is a physiologically motivated design that failed cleanly. Decoding from the 0.1–3 Hz onset-locked band alone gives 0.604; from the mu and beta ERD band alone, 0.775; combining the two, 0.742. The paired difference between the combination and the better single band is -0.034 , so on single-trial data the low-frequency pathway adds noise, and the architecture uses the ERD band only.

6.7. A rejected explanation for the inversion

One candidate explanation is the power-weighted estimator described in Section 4.1. The defect is real and measurable: on cohort B, where the majority class carries $1.52\times$ the power, the un-normalised running estimate sat at Riemannian distance 0.306 from the walk covariance, which is to say it had become the majority class. The per-window trace normalisation in (3) moves it to 3.899 and 3.194 from the two class covariances.

Correcting it changed decoding by $+0.005$ on cohort B and -0.016 on cohort A. The estimator improved and the task did not follow, so this is therefore not the explanation for the inversion, though the correction is retained, which is independently right. It is the first of five cases in this work where a distribution-level quantity improved without the task following (Section 7).

6.8. The adaptation rate condition

The running estimate tracks whatever is currently streaming. An exponential moving average with momentum m has an effective memory of approximately $1/m$ batches, or NB/m windows at batch size N and B batches. When a protocol presents a class in contiguous blocks longer than that memory, the estimate converges to the currently streaming class, and whitening by it removes the between-class difference. Cohort A's blocks ($\tau_{\text{blk}} = 18$) sit far inside even the fastest memory; cohort B's ($\tau_{\text{blk}} = 309$) do not.

If this mechanism is responsible, slowing adaptation until the memory exceeds a class block should restore cohort B. The memory is approximately 160, 640 and 3200 windows

at $m = 0.2, 0.05$ and 0.01 , so the crossing of $\tau_{\text{blk}} = 309$ falls between the first and second settings.

Every arm that uses alignment gains $+0.090$ to $+0.098$ as the memory crosses the class-block length (Table 4). Neither arm that omits alignment responds: the no-align arm moves -0.010 , within its 0.0107 run-to-run spread (Table 9), and the stem-only arm reproduces 0.7375 exactly at every setting despite a 160-fold change in the adaptation rate. Because that arm is deterministic, this null is exact. The manipulated variable therefore acts only through the component it controls, which an unrelated confound such as optimisation dynamics or regularisation would not produce.

6.9. The condition is two-sided

On cohort A, whose blocks sit far inside even the fastest memory so that the estimate is already class-mixed, the same change was expected to be neutral. This expectation was registered before the experiment was run. It is not borne out (Table 4).

The optima are opposite, and error in either direction costs 0.060 to 0.092, about 2.3 to 3.6 times the standard deviation across data-split seeds. Slower adaptation therefore costs accuracy on cohort A, indicating a trade-off: a fast estimate tracks both the streaming class, which is harmful, and genuine session drift, which is beneficial.

That cost was first measured on *align + ctx + gate*, which is not the configuration this paper proposes. Repeating it on the proposed *align + gate* arm enlarges the figure: slowing to $m = 0.01$ takes cohort A from 0.884 to 0.799 ± 0.087 (SD across the seven participants), a cost of -0.085 under the trace-normalised estimator. Both numbers use the same estimator version; comparing the slow arm against the original-estimator 0.901 would inflate the cost to -0.102 by mixing estimator versions. The comparison is unpaired, since per-participant values were not retained for the $m = 0.2$ reference, and inferential testing was therefore not performed on it.

Condition 1 (Admissible rate band). Assume (i) the running estimate is an exponential moving average with effective memory μ windows; (ii) the within-class covariance is stationary over a block; (iii) the protocol presents classes in contiguous blocks of length τ_{blk} ; and (iv) session drift acts as a congruence transform slow on the scale τ_{drift} . Under (i)–(iv) the estimate both avoids tracking the streaming class and tracks session drift only if

$$\tau_{\text{blk}} < \mu < \tau_{\text{drift}}. \quad (4)$$

Where $\tau_{\text{blk}} \geq \tau_{\text{drift}}$ the band is empty and no setting of μ satisfies both requirements. It is a design condition that states when the layer can work. It is established here empirically; we give no proof.

Cohort A satisfies (4) with a wide margin, so fast adaptation is both safe and useful there, and slowing it forfeits the drift tracking for no compensating gain. Cohort B's 309-window blocks force μ slow enough that it can no longer track drift, so the band is empty.

Table 13

Predictions registered before any model was trained on cohort C. Balanced accuracy averaged over all 20 participants, seed 0. Differences are paired within participant.

Prediction	Arms compared	Difference	Threshold	Outcome
P1 alignment costs little	0.783 / 0.785	+0.002	> -0.05	holds
P2 slowing does not help	0.785 / 0.779	-0.006	< +0.05	holds

Remark 1 (The residual). This accounts for the residual without a third mechanism. Even at its best setting, alignment still costs 0.060 on cohort B (0.723 against 0.783 for the no-align arm), because at the only rate that avoids class tracking there is no drift-tracking benefit left to collect. The layer then contributes only estimation noise.

Remark 2 (Offline alignment is exempt). A whole-recording estimate, as in Euclidean Alignment [13], is the maximally slow setting. It therefore never tracks the class, and it never tracks within-recording drift either, which is exactly the capability an in-network version is introduced to add.

Both bounds in (4) are properties of the recording protocol. τ_{blk} is read off the trial structure. τ_{drift} can in principle be estimated without labels from the Riemannian distance between covariances over time, but a block-wise estimate taken within a session is dominated by class alternation wherever blocks are short, and that measurement is not reported here (Section 8). The lower bound is therefore checkable before a model is fitted, and Section 6.10 tests it on a cohort whose results were not available when the prediction was made.

6.10. A prediction registered before a third cohort

Condition 1 was formulated after cohort B failed, so on the two cohorts above it is an explanation written with their results in view. To test it in advance, predictions for a third cohort were committed to version control before any model was trained on it. The cohort, participants, data split, preprocessing, compared arms and decision thresholds were all fixed at that point.

Cohort C alternates rest and movement about every 4 s. From the published protocol alone a class block holds about five windows, and the loader later measured a median of 5.0. Against an estimator memory of 160 windows at $m = 0.2$, the lower bound of (4) is satisfied by a factor of about thirty, so the estimate cannot converge to either class. Two predictions follow. P1: enabling alignment costs less than 0.05, where the same comparison cost 0.211 on cohort B. P2: slowing adaptation to $m = 0.01$ gains less than 0.05, where it recovered 0.090 on cohort B. The arms compared in P1 differ only in alignment, and those in P2 only in momentum.

Both predictions hold (Table 13). Enabling alignment changes balanced accuracy by +0.002 (higher in 11 participants, lower in 9; Wilcoxon $p = 0.68$), and slowing it changes accuracy by -0.006 (higher in 9, lower in 10; $p = 0.43$). The intervention that recovered 0.090 on cohort B brings no gain here, as the protocol predicted, and enabling alignment shows no harm.

Two further seeds were run after the registered test and are reported separately from it. They agree with both verdicts and sharpen one of them. Enabling alignment changes accuracy by +0.014 (higher in 13 participants, lower in 7; $p = 0.17$), so it again costs nothing, and its small positive trend is not significant. Slowing adaptation costs 0.016 (lower in 15 of 20; $p = 0.003$) and is negative on all three seeds, so on this cohort it carries a small cost. Read after the fact, the sign of slowing now fits the band on every cohort. It helps where the estimate tracks the class (cohort B, +0.090), costs most where the estimate was tracking drift across weeks (cohort A, -0.085), and costs least where drift is confined to a single session (cohort C). That ordering was observed on the replication seeds and formed no part of the registered prediction.

The test covers the short-block side of Condition 1 and its contrast with cohort B. It does not test an empty band, which requires a protocol whose blocks outlast its drift.

7. Discussion

7.1. Validating a mechanism separately from performance

Domain adaptation and invariance methods are routinely justified by a distribution-level quantity, a divergence, an alignment distance or a nuisance-decodability score, with task performance assumed to follow [71–74]. This work provides five counterexamples within a single system, three of which we produced by acting on that assumption ourselves (Fig. 7).

1. The adaptation rate was originally selected because it maximised drift reduction (60 % against 25 %). That choice cost 0.092 accuracy on cohort B (Table 4).
2. The cohort in which more drift was removed (83.9 % against 52.0 %, Table 8) is the cohort in which accuracy fell by 0.166 (Table 12).
3. A verified correction to a genuine covariance-estimation bias moved decoding by +0.005 and -0.016 (Section 6.7).
4. A prediction registered from our own diagnosis was falsified (Table 4), because the diagnosis accounted only for the harm the estimate can do.
5. The nuisance-leakage probe rises with accuracy whether or not the decoder can access the nuisance, measuring $R^2 = +0.863$ for one that cannot.

Item 4 generalises the rest. Each of these quantities is one-sided by construction: it scores the failure it was designed to detect and carries no term for the capability the intervention removes. Drift reduction measures how much non-stationarity a layer removes and has no term for what the layer destroys. A diagnosis inherited from such a metric inherits its blind spot, which is why our own threshold rule was correct about one direction and silent about the other until we tested it where it predicted no effect. The practical implication is that reporting a reduced divergence, increased invariance or lowered nuisance decodability is evidence

about the representation alone, saying nothing about whether the method helps, even when the reduction is large and reproducible.

7.2. Relation to temporally correlated test-time adaptation

The vision literature has identified that test-time adaptation degrades when the test stream is temporally correlated [20–22], and proposes reservoirs and stabilised objectives as remedies. Our result is compatible with that line and adds two things specific to the blocked-protocol case. First, the relevant quantity is a ratio of timescales, measured on the stream, and which side of the threshold a protocol falls on can be predicted in advance, as the third cohort shows (Section 6.10). Second, the constraint is two-sided, because in a longitudinal deployment the same estimator that can track the class is also the one that tracks the drift, so the remedies that slow adaptation trade one failure for the other instead of removing both.

8. Limitations

Several limitations constrain the interpretation of these results.

No published baseline on cohort B. Every decoder in Table 3 was run on cohort A. Nothing from the literature has been run on cohort B in this pipeline, so the 0.792 reported there can be compared with our own arms and with the majority-class floor, but not with the field. The decoder therefore transfers to cohort B; its standing relative to published methods there is untested.

Seeds. Most ablations are seed 0. The proposed configuration was rerun over three data-split seeds under the trace-normalised estimator, giving 0.868 ± 0.026 (seeds 0.884, 0.882 and 0.839). That seed standard deviation is larger than the 0.010 noise floor of repeated runs at a fixed split (Table 9), and it is concentrated in two participants, whose accuracy ranges over 0.217 and 0.120 across seeds while the other five vary by at most 0.065. Single-seed rankings of closely spaced models are therefore not interpretable, so the 0.012 gap to ATCNet in Table 3 is reported as a loss and its direction is left unresolved. The effects we do interpret (+0.090 to +0.098, −0.060) are about 2.3 to 3.8 times the seed standard deviation, a narrower margin than the fixed-split noise floor suggests. The paired alignment effect on cohort A over the same three seeds is +0.026 ($p = 0.22$) and is negative on one seed, so the benefit of alignment where the band holds is small and unconfirmed at this sample size. The loss where the band fails, 0.211 on cohort B, comes from a single seed and has not been repeated.

No paired significance tests on accuracy. Per-participant accuracies were not retained for the component ablation and the rate tables, so those comparisons carry dispersion without paired tests. Paired tests are reported wherever per-participant rows exist: the drift results (Table 8), the temporal and normalisation controls, the artefact control (Section 6.3), the three-seed alignment

comparison on cohort A (Section 6.2) and the cohort C predictions (Section 6.10).

Cohort size. Seven, eight and twenty participants. This is typical for exoskeleton BCI studies [1] but small, and the between-subject SD of 0.06 to 0.13 (Table 10) is large relative to the differences between arms.

Best-epoch evaluation. Model selection uses the best epoch by held-out balanced accuracy, which is conventional but upward-biased on small folds. The bias applies equally to all arms and baselines within a pipeline, so comparisons within a pipeline are unaffected, but absolute values should be read as optimistic.

The upper bound is only bracketed. τ_{drift} in (4) is bounded by cohort A performing well at a 160-window memory and poorly at 3200. We have not measured an interior optimum, and a cohort whose admissible band is narrow but non-empty would test Condition 1 properly. A direct estimate of τ_{drift} from block-wise covariance distances within a session is dominated by class alternation where blocks are short. On cohort A it returns about 18 windows, which is the class-block length, so it is not reported.

Condition 1 is hypothesis-generating. It was formulated after observing cohort B fail. One prediction from it has since been tested out-of-sample, and that test falsified the one-sided form (Table 4), which is how the two-sided form arose. The corrected form has since been tested prospectively on cohort C (Section 6.10), where both registered predictions held. That test covers the lower bound and the contrast with cohort B; the upper bound has not been tested in advance.

Artefact. Retraining after ICA cleaning lowers accuracy by 0.051, concentrated in three participants (Section 6.3). The losses do not follow the amount of muscle activity removed. With seven participants, and with re-referencing part of the cleaning, the control still falls short of establishing that the accuracy is neural. ICLabel is also not trained on gait-related artefact specifically.

Window length differs between cohorts. Cohort B uses 2 s windows against cohort A's 4 s, so the context pathway spans a shorter duration there. This is a confound for the context ablation specifically, though not for the rate condition, which is measured within each cohort.

9. Conclusion

An estimate that adapts at test time tracks whatever is currently streaming. Under a protocol that presents each class in long contiguous blocks, the estimate converges to the current class, and the layer then removes the signal it was inserted to protect. The estimator's memory must outlast a class block and stay shorter than the drift timescale, and where those requirements cross no admissible rate exists. Both bounds are read from the recording protocol, which makes this a design-time check.

The condition has predicted a new cohort correctly in advance. With class blocks of five windows, enabling alignment cost nothing and slowing it brought no gain, where the same slowing recovered 0.090 on the cohort with long

blocks. Where the band holds, the gain from alignment is small and not significant on either cohort that satisfies it: +0.026 over three seeds on cohort A and +0.014 over two replication seeds on cohort C. Where the band fails, alignment costs 0.211. The condition is therefore most useful as a design-time guard against that loss.

The immediate open question is whether the band in (4) has an interior optimum that can be located exactly. No cohort here can answer it. Cohort A's band is wide enough that the fastest setting tested is already inside it, cohort C's is wider still, and cohort B's is empty. A protocol with τ_{blk} between roughly 100 and 200 windows would place the two bounds close together, and the prediction is that accuracy would peak at an intermediate momentum and fall on both sides. That experiment can be pre-registered from the protocol alone, as the cohort C test was, and it would test the upper bound, which no cohort here has tested in advance.

A second direction follows from the residual. If a single running estimate cannot simultaneously avoid the class and track the drift, then the two requirements should perhaps be served by two estimators at different rates, with the fast one constrained to the subspace in which drift lives and the slow one supplying the class-neutral reference. Whether such a decomposition is identifiable without labels remains open.

These observations may extend to other online adaptation settings. In five cases here a distribution-level quantity improved without a corresponding task improvement, and in three of those the assumption that it would had already been acted upon. Such measures are one-sided by construction, which suggests that methods justified by a reduced divergence should also report the accuracy cost of that reduction.

CRedit authorship contribution statement

Esm E Moula Chowdhury Abha: Software, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Md Wahiduzzaman Suva:** Software, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Muhammad Hasibur Rashid Chayon:** Supervision, Project administration, Writing – review and editing.

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Both datasets are publicly available. All code, configurations and result files, including those of experiments that failed, are in the accompanying repository.

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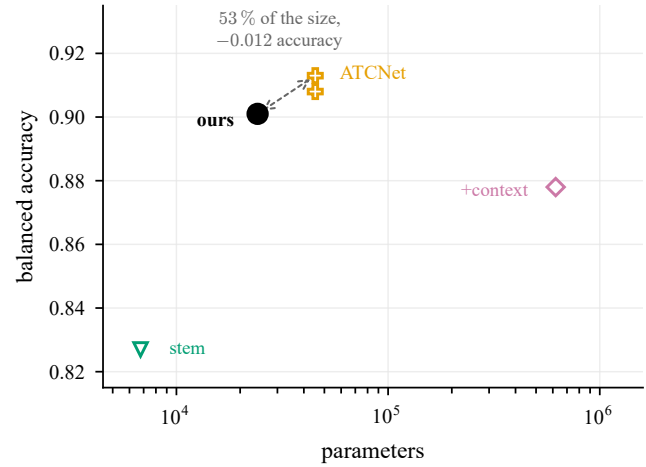


Figure 4: Balanced accuracy against parameter count on cohort A, single seed, pipeline C only. The proposed configuration sits 0.012 below ATCNet at 53 % of its parameters. No point from the other pipeline appears here.

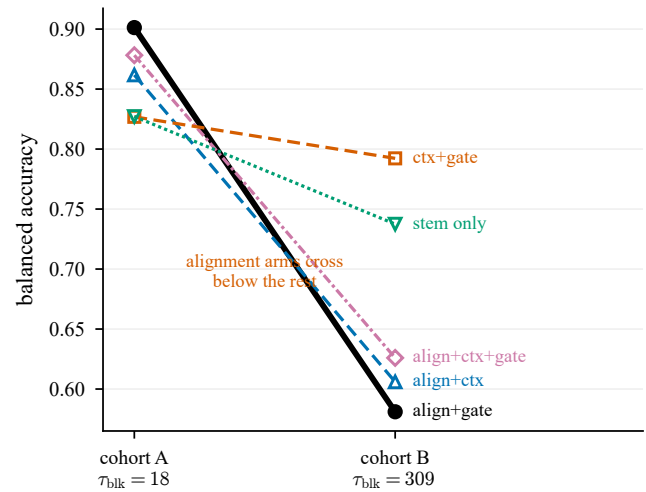


Figure 5: The same five arms on both cohorts, single seed. Solid heavy black is the proposed configuration; the other alignment-using arms are dashed in colour, and the two alignment-free arms are dashed and dotted. The three alignment arms cross below both alignment-free arms, so the ordering inverts instead of compressing.

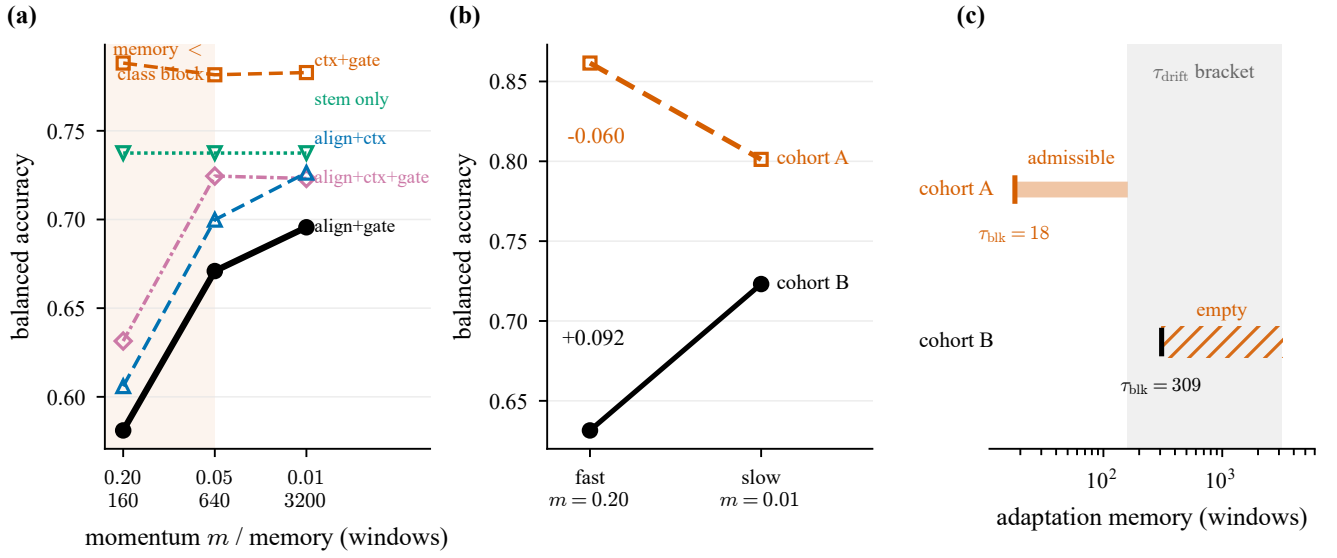


Figure 6: The adaptation rate condition. (a) Cohort B, five arms, three momenta, single seed; the shaded region is where the estimator memory is shorter than a class block. Measurement points are marked because the sweep is sparse. (b) The same change applied to both cohorts at matched estimator version, giving opposite signs. (c) The admissible band of Condition 1, bounded below by τ_{blk} and above by τ_{drift} ; the upper bound is bracketed by cohort A being good at 160 windows and poor at 3200.

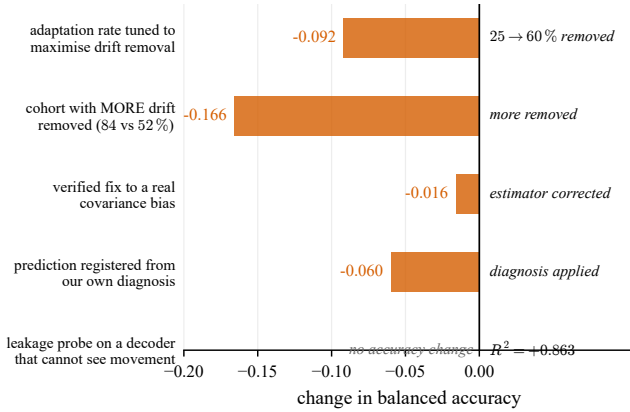


Figure 7: Five cases in which a distribution-level quantity improved (right, in blue) and the task did not follow (left, in red). The final row has no accuracy change and therefore no bar: the leakage probe rises with accuracy whether or not the decoder can access the nuisance.